CORTEXIS SOLUTION HUB

Comprehensive Tools & Converters Roadmap

Master catalog and architectural blueprint for expanding the Cortexis Free Online Toolkit ahead of the end-of-month production deployment.

CURRENT TOOLKIT	PROPOSED ADDITIONS	TARGET ARCHITECTURE	TARGET DEPLOYMENT
62 Implemented Tools	**+42 High-Value Utilities**	**100% Client-Side / WASM**	**End of Month 2026**

+7	+16	+8	+11
Essential PDF Tools	Missing Converters	Image & Graphics Tools	Developer & Text Tools

> **Strategic Advantage:** Cortexis leverages client-side WebAssembly, Canvas, and pure JS engines (`pdf-lib`, `tesseract.js`, `ffmpegEngine.js`). Adding these recommended utilities does *not* increase server RAM or cloud compute overhead, ensuring infinite scalability and zero privacy liability.

P 1. High-Demand Missing PDF Tools

Critical additions for business, academic & legal workflows

TOOL NAME	UTILITY & KEY FEATURES	LIBRARY / ENGINE	PRIORITY
PDF Watermark Adder	Add text (Confidential, Draft) or PNG/JPG logo stamps to all or selected pages with opacity, angle, and position controls.	`pdf-lib`	TIER 1 · URGENT
Add Page Numbers to PDF	Insert page numbers ("Page 1 of N", Arabic, Roman) with custom position, margins, fonts, and starting number.	`pdf-lib`	TIER 1 · URGENT
PDF e-Sign (Sign PDF)	Draw signature on canvas, type signature, or upload stamp image; drag-and-drop to position on any page and export.	`HTML5 Canvas + pdf-lib`	TIER 1 · URGENT
PDF Redact / Blackout	Permanently black out sensitive data (CNIC, tax IDs, credit cards) so confidential information cannot be highlighted or extracted.	`pdf-lib (DrawRect)`	TIER 2
PDF to Grayscale (B&W)	Convert color PDFs to pure grayscale to reduce print ink costs and meet government portal upload guidelines.	`pdfjs-dist + Canvas`	TIER 2
Compare PDFs (Visual Diff)	Side-by-side comparison of two revisions of a document, highlighting added/removed text and visual differences.	`pdfjs-dist + Pixelmatch`	TIER 3
PDF Bookmarks / TOC Builder	Add or edit PDF bookmarks, outlines, and interactive table of contents for long documents and ebooks.	`pdf-lib Outlines`	TIER 3

C 2. Essential Missing Converters (Document, Media & Data)

Highest Google search volume queries in the tools industry

CONVERTER NAME	TRANSFORMATION & PURPOSE	TECHNOLOGY	PRIORITY
Word to PDF (DOCX → PDF)	Direct conversion of Word documents into styled, printable PDFs without requiring Microsoft Office.	`mammoth + jspdf`	TIER 1 · URGENT
Excel to PDF (XLSX → PDF)	Convert spreadsheets and tables into formatted, page-aligned PDF documents.	`xlsx + jspdf-autotable`	TIER 1 · URGENT
Image to PDF (JPG/PNG → PDF)	Combine multiple receipts, photos, or scans into a single, clean multi-page PDF document.	`pdf-lib`	TIER 1 · URGENT
HEIC / HEIF to JPG/PNG	Convert iPhone photos (HEIC) to universal JPG/PNG formats. (Massive search traffic on Windows/Android).	`heic2any (Client WASM)`	TIER 1 · URGENT
AVIF to JPG / PNG / WebP	Convert modern AVIF images to standard formats for legacy browser and editor compatibility.	`HTML5 Canvas ImageBitmap`	TIER 2
GIF to MP4 Converter	Compress heavy 20MB animated GIFs into 500KB ultra-fast MP4 videos for web embedding.	`ffmpegEngine.js`	TIER 1 · URGENT
WEBM to MP4 / MP4 to WEBM	Convert browser screen recordings and OBS clips into universally playable MP4 files and vice-versa.	`ffmpegEngine.js`	TIER 2
OGG / OPUS to MP3	Convert WhatsApp voice notes (.opus / .ogg) and Telegram audio into universally playable MP3 files.	`ffmpegEngine.js`	TIER 1 · URGENT
M4A / AAC to MP3	Convert Apple voice memos and iTunes audio files to standard MP3.	`ffmpegEngine.js`	TIER 2
CSV ↔ JSON Converter	Two-way conversion between tabular CSV data and nested/array JSON objects with instant download.	`PapaParse / Native JS`	TIER 1 · URGENT
XML ↔ JSON Converter	Convert legacy XML documents to clean JSON and vice-versa for API debugging.	`fast-xml-parser`	TIER 2
YAML ↔ JSON Converter	Convert Kubernetes / Docker-Compose YAML configurations into JSON and vice-versa.	`yaml package`	TIER 2
SQL to CSV / CSV to SQL	Convert SQL dump results to spreadsheet CSV, or CSV tables into bulk SQL `INSERT INTO` statements.	`Native JS Parser`	TIER 2
Markdown to PDF / HTML	Live markdown editor that compiles and prints clean documentation with syntax-highlighted code.	`react-markdown + jspdf`	TIER 2
RAR / 7Z to ZIP Converter	Extract or re-pack uncompressed/proprietary archives into standard cross-platform ZIP files.	`libarchive.js (WASM)`	TIER 3

CONVERTER NAME	TRANSFORMATION & PURPOSE	TECHNOLOGY	PRIORITY
Unix Timestamp to Date	Convert epoch milliseconds/seconds to human dates across all timezones and vice-versa.	`Native Date API`	TIER 1 · URGENT

3. Missing High-Demand Image & Graphics Tools

E-commerce, privacy and design utilities

TOOL NAME	UTILITY & KEY FEATURES	TECHNOLOGY	PRIORITY
AI Background Remover	1-click background removal for product photos and portraits entirely inside the browser using WebAssembly.	`@imgly/bg-removal`	TIER 1 · HIGH
Favicon & App Icon Suite	Upload 1 logo → generates 16x16, 32x32, 180x180 Apple Touch, 192x192, 512x512 PWA icons + `site.webmanifest` zip.	`Canvas + JSZip`	TIER 1 · URGENT
Photo EXIF Privacy Stripper	Read and wipe GPS coordinates, camera serial numbers, and personal metadata from JPEG/PNG photos before uploading.	`exif-js / piexifjs`	TIER 1 · URGENT
Image Watermark & Batch Stamp	Apply custom copyright text or logo image overlays onto photos in bulk with opacity and position controls.	`HTML5 Canvas`	TIER 2
Blur / Censor / Pixelate Tool	Easily blur or pixelate faces, car license plates, and sensitive documents with a brush or rectangle tool.	`Canvas Image Filters`	TIER 2
Image OCR (Image to Text)	Standalone dedicated tool to extract text from screenshots, invoices, and photos. (`tesseract.js` already installed).	`tesseract.js`	TIER 1 · URGENT
SVG Optimizer & Cleaner	Minify SVG code, strip editor metadata (Illustrator/Figma artifacts), and clean path data for faster web load.	`svgo in browser`	TIER 3
Meme Generator	Classic meme templates + custom upload with editable top/bottom impact captions and instant PNG export.	`Canvas Context2D`	TIER 2

D 4. Developer & Code Utilities (DevToys / CyberChef Style)

Essential for developers, devops engineers, and QA teams

TOOL NAME	UTILITY & KEY FEATURES	TECHNOLOGY	PRIORITY
JWT Decoder & Inspector	Decode JWT Header, Payload, Claims, and Expiration timestamps in browser without exposing secret tokens over internet.	`Native Base64URL`	TIER 1 · URGENT
Hash & Checksum Generator	Calculate MD5, SHA-1, SHA-256, and SHA-512 hashes for pasted text or uploaded files for integrity checks.	`Web Crypto API`	TIER 1 · URGENT
UUID / GUID & Secret Generator	Generate batch v4 UUIDs, NanoIDs, cryptographic random hex tokens, and strong passwords.	`crypto.randomUUID()`	TIER 1 · URGENT
SQL Formatter & Beautifier	Beautify messy SQL queries for MySQL, PostgreSQL, SQLite, and Oracle with indentation and keyword capitalization.	`sql-formatter`	TIER 2
CSS Visual Generator Suite	Interactive visual UI generators for: (1) Multi-layer Box Shadows, (2) Glassmorphism, (3) Mesh & Linear Gradients.	`React + CSS Variables`	TIER 2
RegEx Tester & Explainer	Real-time regular expression evaluation with color-coded capture groups and cheat-sheet library.	`RegExp JS Engine`	TIER 2
Text & Code Diff Checker	Side-by-side visual difference highlighter showing additions, deletions, and line changes between two files.	`diff / jsdifflib`	TIER 1 · URGENT
URL Encoder / Decoder	Clean component URL encoder, decoder, and query string parameter parser.	`encodeURIComponent`	TIER 1 · URGENT

A 5. Missing Text, Video & Audio Quick Tools

Streamlining everyday content creation and media editing

TOOL NAME	UTILITY & KEY FEATURES	TECHNOLOGY	PRIORITY
Word & Character Counter	Live word count, character count, sentence count, reading time (WPM), speaking time, and keyword density.	`Pure React Engine`	TIER 1 · URGENT
Case Converter Suite	Convert text to UPPERCASE, lowercase, Title Case, camelCase, snake_case, kebab-case, and Sentence case.	`String Regex`	TIER 1 · URGENT
Duplicate Line Remover & Sorter	Clean up email lists, IDs, and raw text by stripping duplicates, blank lines, and sorting alphabetically.	`Set / Array Sort`	TIER 1 · URGENT
Video Trimmer / Cutter	Slice start and end times of MP4/MOV videos with visual scrub slider and export trimmed video.	`ffmpegEngine.js`	TIER 1 · URGENT
Audio Trimmer / Ringtone Maker	Trim audio tracks with waveform preview to create custom ringtones and clips.	`Web Audio API / FFmpeg`	TIER 2
Video Audio Remover (Muter)	Strip audio track completely from video clips in 2 seconds.	`ffmpegEngine.js (-an)`	TIER 1 · URGENT
Video Speed Adjuster	Speed up or slow down videos (0.5x, 0.75x, 1.25x, 1.5x, 2x) preserving audio pitch.	`ffmpegEngine.js`	TIER 2

R 6. Implementation Roadmap & Production Deployment Checklist

Timeline leading to end-of-month deployment

PHASE	DELIVERABLES & SCOPE	STATUS / TARGET
Phase 1 (Days 1–3) Fast High-Value Wins	• PDF Watermark, Page Numberer, and e-Signature Tool (using existing `pdf-lib`). • Word to PDF & Image to PDF converters. • JWT Decoder, Hash Generator, UUID Generator, and Word/Case Converters. • Favicon & App Icon Suite.	**Ready for immediate build**
Phase 2 (Days 4–7) Media & Converters	• Video Trimmer, Audio Muter, and OGG-to-MP3 converter (using `ffmpegEngine.js`). • CSV ↔ JSON, XML ↔ JSON, and Timestamp converters. • Standalone Image OCR (using existing `tesseract.js`). • Text Diff Checker and SQL Beautifier.	**Scheduled for mid-sprint**
Phase 3 (Days 8–10) Advanced Features	• AI Background Remover (client-side WASM). • CSS Generator Suite (Shadows, Gradients, Glassmorphism). • Photo EXIF Privacy Stripper & Censor/Blur Tool. • Favorites & "Recently Used" LocalStorage shelf on `/tools` page.	**Scheduled pre-deployment**

Production Deployment Technical Checklist:

CHECKPOINT	ACTION REQUIRED	DEPLOYMENT RISK
Docker Chromium Setup	`Dockerfile` uses Alpine for `npm run prerender`. Must ensure `apk add chromium` & `ENV CHROME_PATH=/usr/bin/chromium-browser` are set.	**High (Build Failure)**
PostgreSQL Data Volume	Ensure `cortexis_pg_data` volume persistence and configure automated daily `pg_dump` cron backup on the host.	**Medium (Data Loss)**
Redis & BullMQ Durability	Redis password & volatile-lru policy verified in `docker-compose.yml`. Ensure email worker stays healthy.	**Low (Verified)**
Client Error Boundaries	Wrap tool view with `<ErrorBoundary>` to catch malformed files gracefully without crashing the React root tree.	**Medium (UX Glitch)**
Sitemap & SEO Sync	Run `node scripts/sitemap.mjs` after registering new tool slugs to update `sitemap.xml` with high SEO rankings.	**Low (Organic Growth)**